



Variations in the responses of North American breeding bird populations to climatic changes

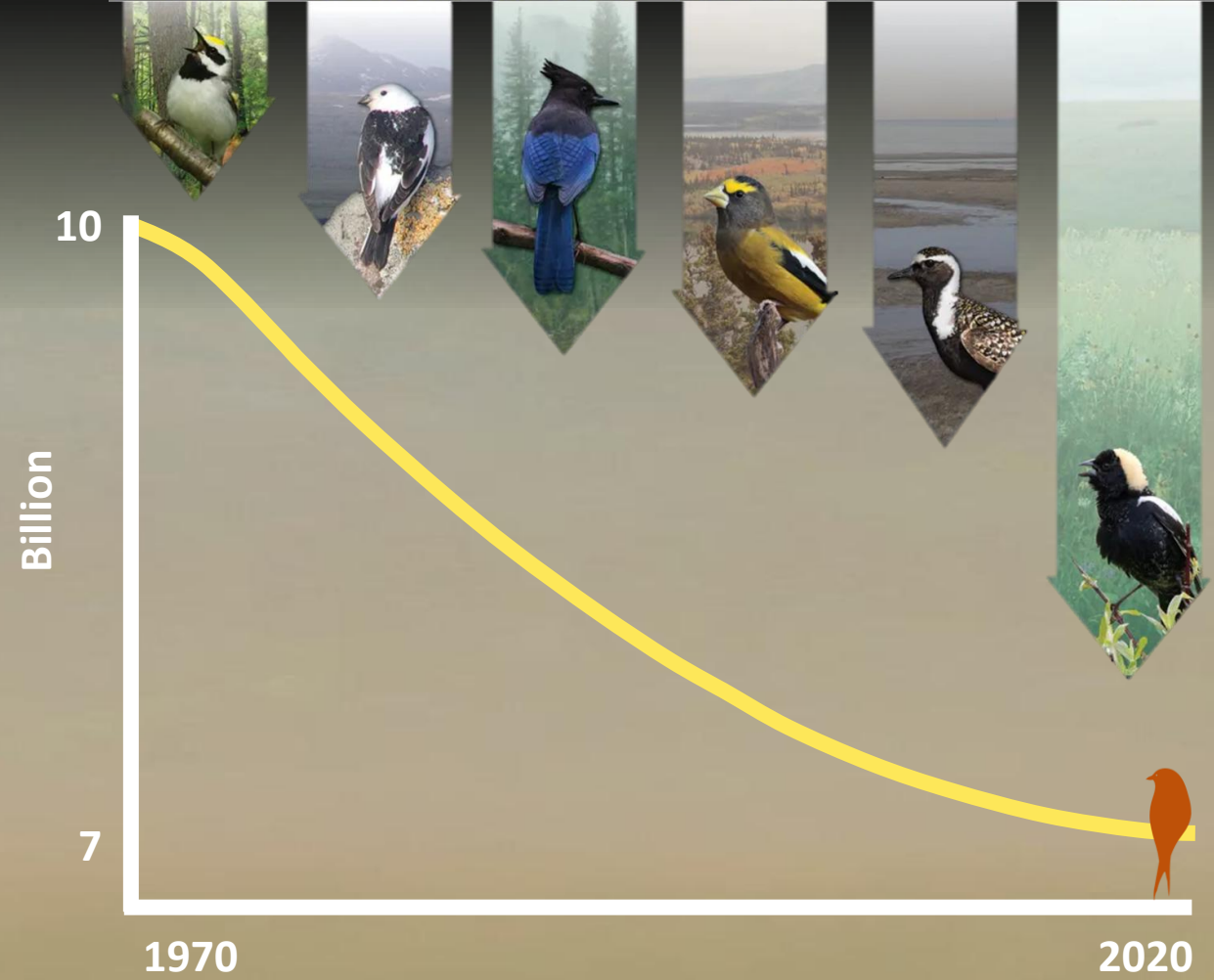
Wentao Song



2.9 billion birds gone since 1970



-17% EASTERN FOREST BIRDS **-23% ARCTIC TUNDRA BIRDS** **-29% WESTERN FOREST BIRDS** **-33% BOREAL FOREST BIRDS** **-37% SHOREBIRDS** **-53% GRASSLAND BIRDS**

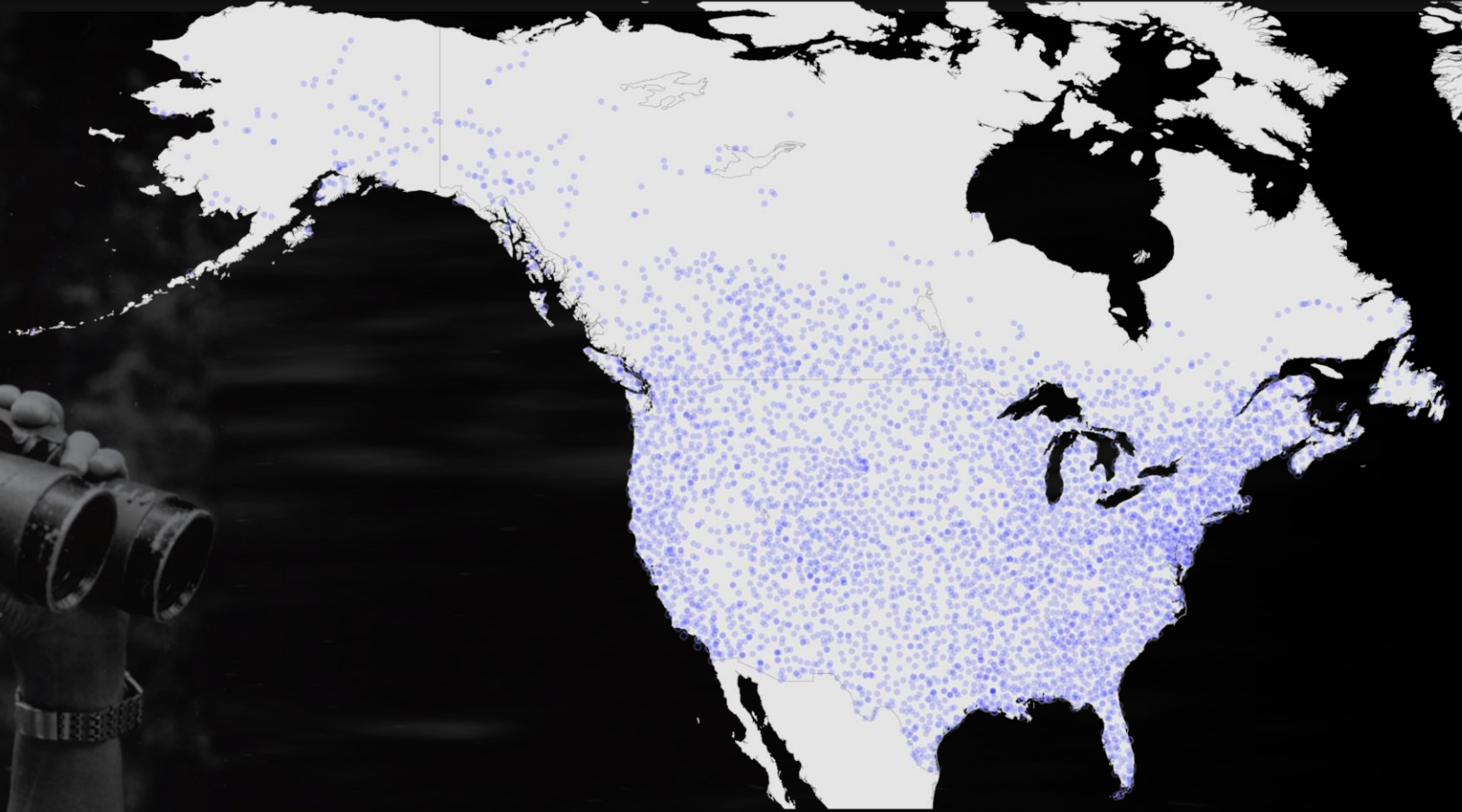


Questions

How and **why** is
North American avifauna
declining?



Breeding Bird Survey since 1966



Chandler S. Robbins

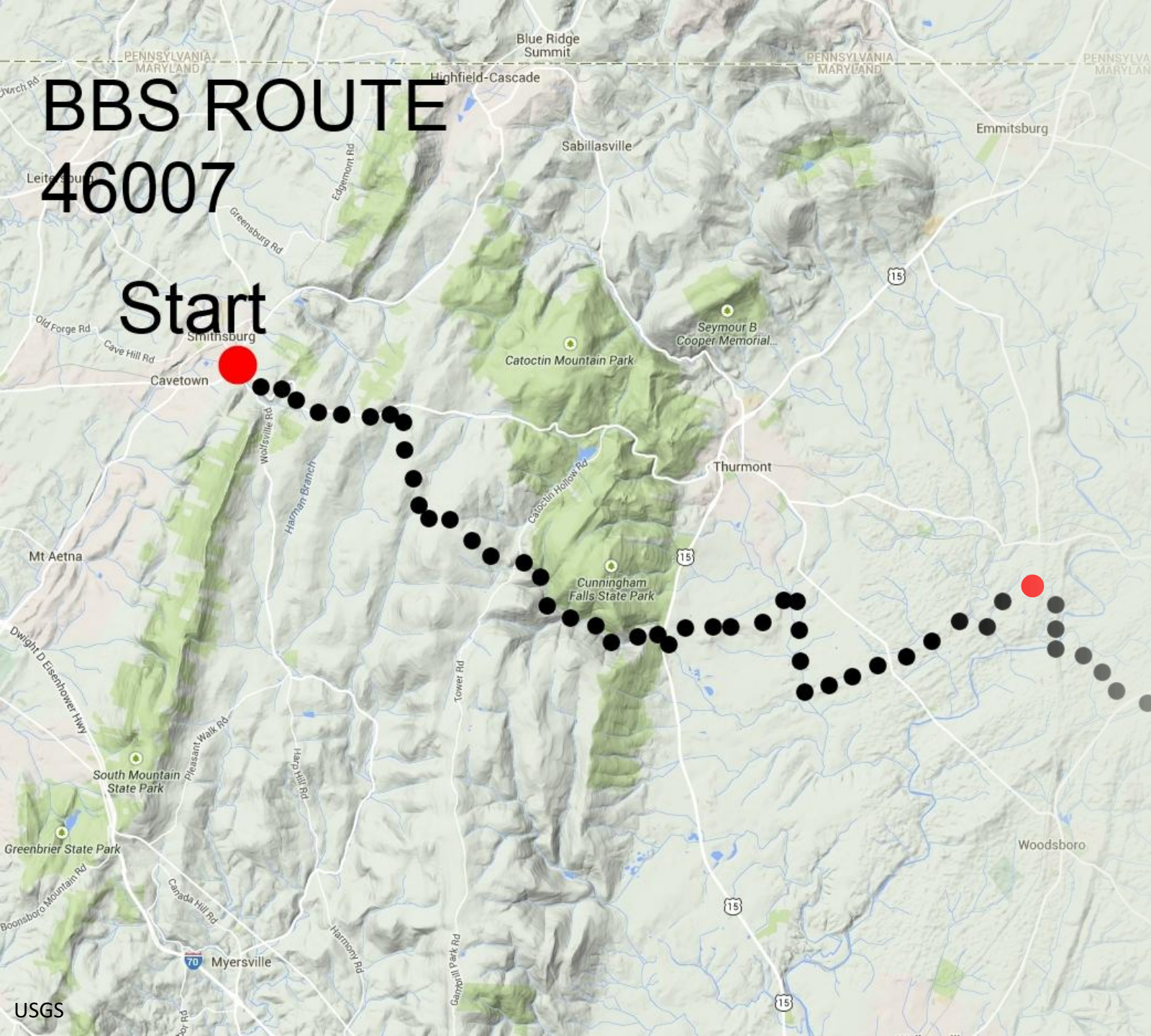
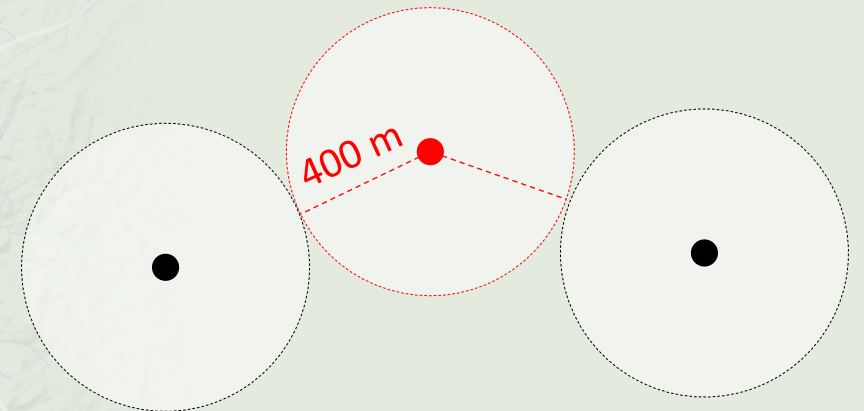
USGS Patuxent Wildlife Research Center

BBS ROUTE 46007

Start

BBS survey protocol

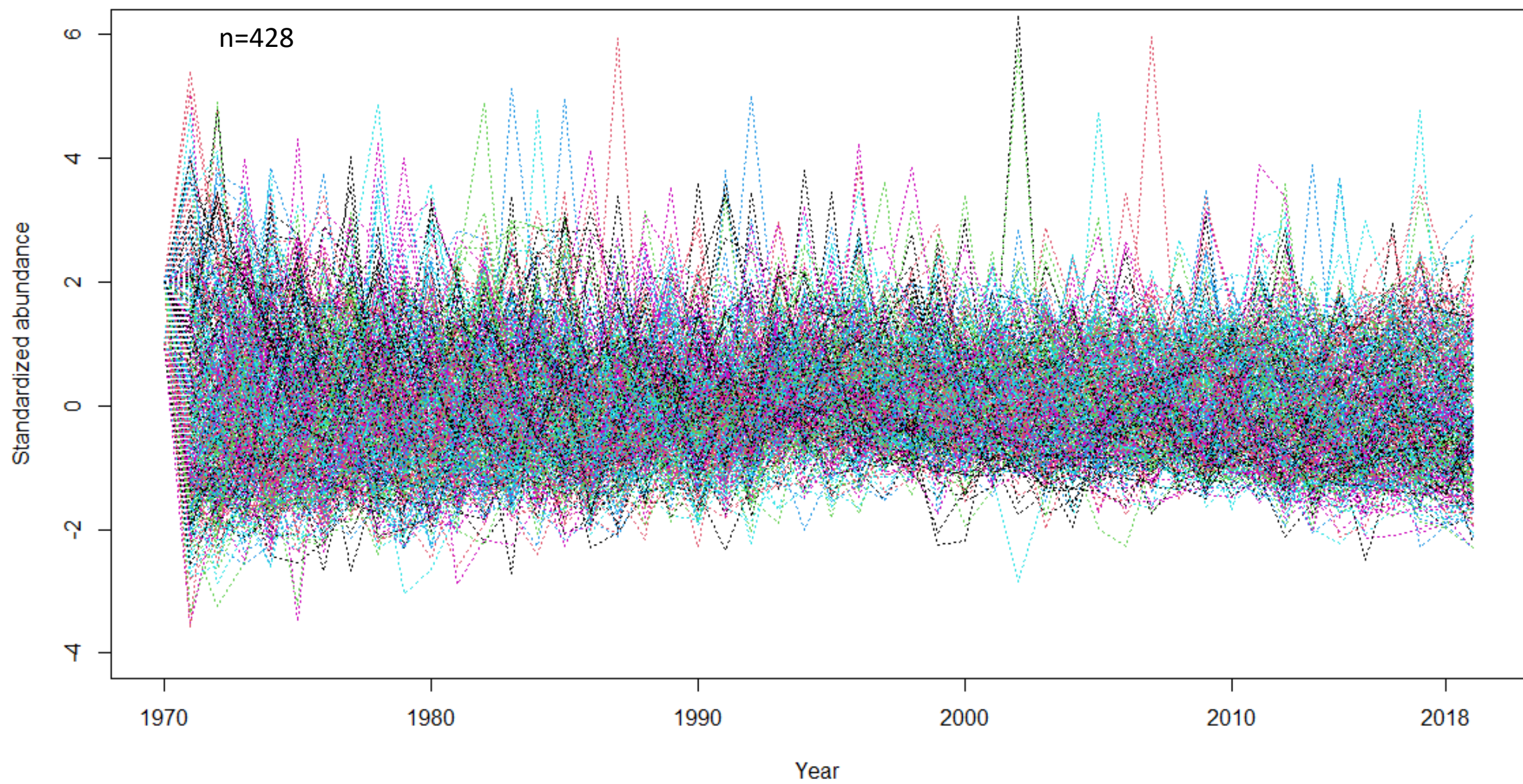
- 40 km
- 50 count spots
- 400 m radius
- 30 min before sunrise
- 3 min



Question 1

How is
North American avifauna
declining?







Sauer et al. 2011

$$\text{Log}(\lambda_{s,j,t}) = S_i + \beta_i(t - t^*) + \omega_j + Y_{i,t} + \varepsilon_{s,j,t}$$

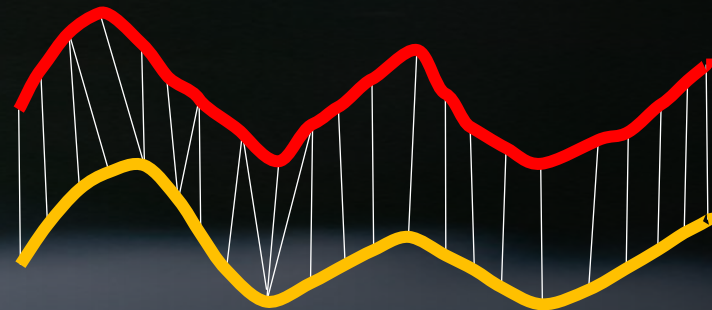
Link et al. 2020

$$Y_s(t) = \alpha_s + \beta_s(t - t^*) + \varepsilon_{st}$$

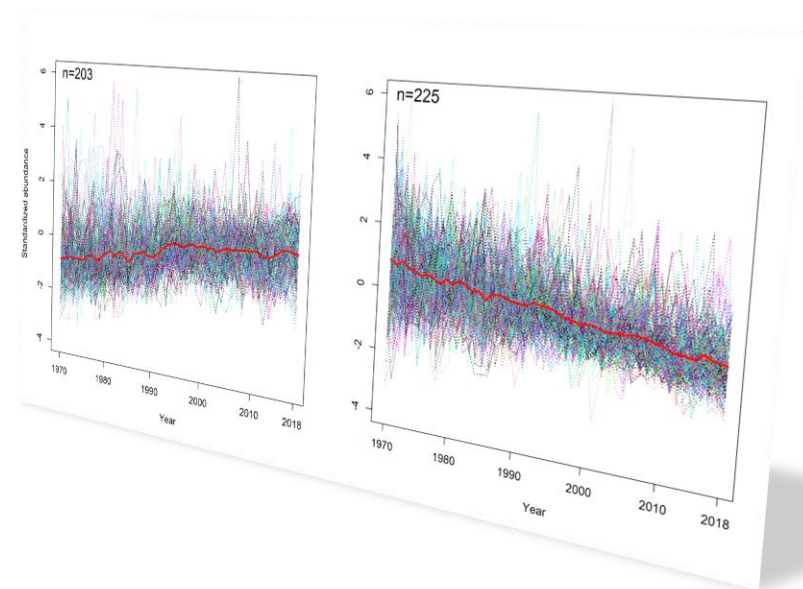
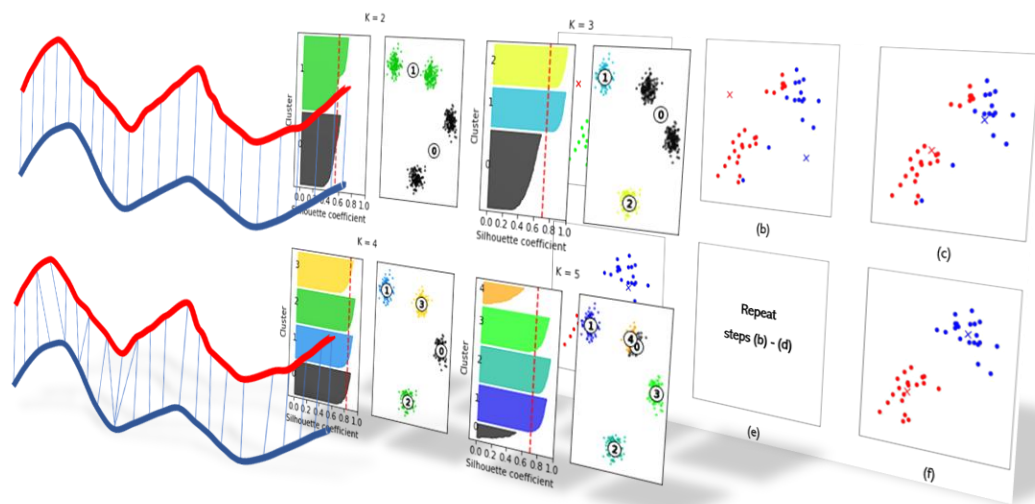
Smith et al. 2021

$$\text{Log}(\lambda_{s,j,t}) = \vartheta_s + \Delta_s(t) + \eta I[j,t] + \omega_j + \varepsilon_{s,j,t}$$

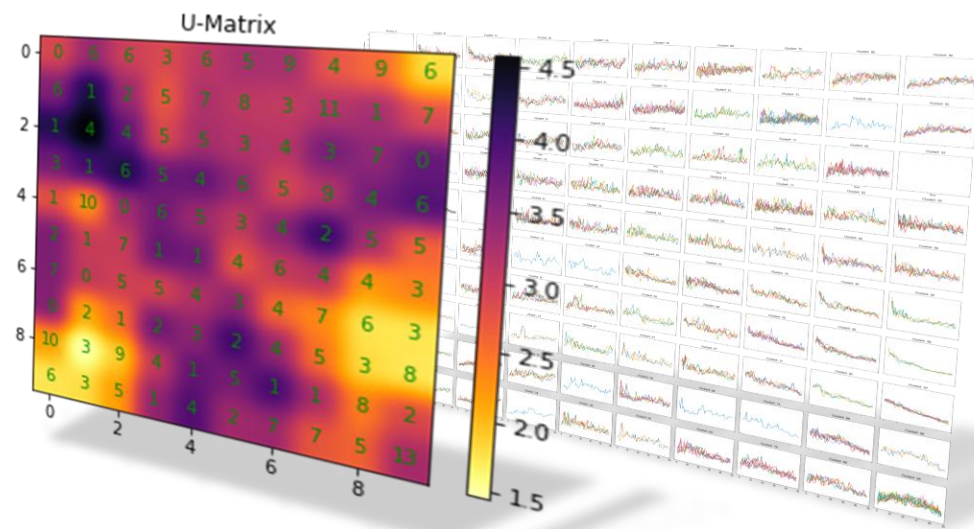
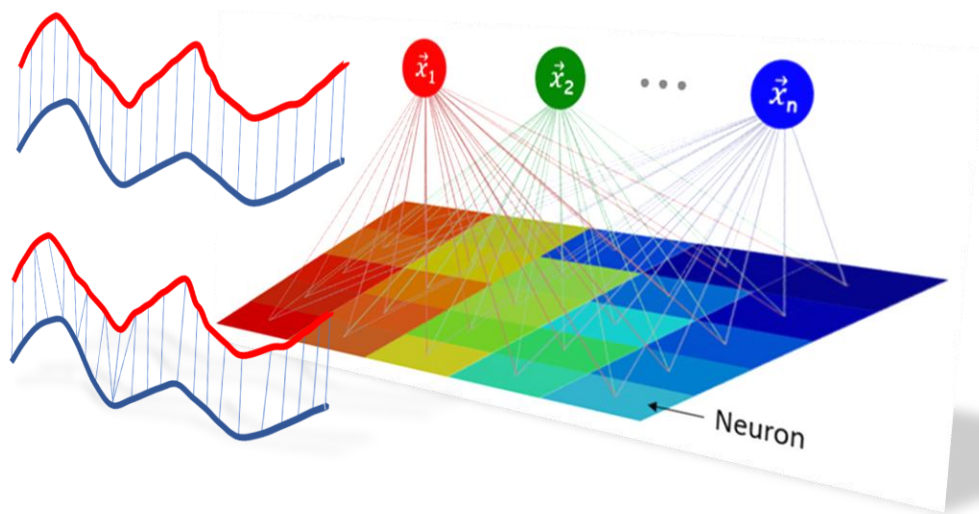
Shape matching?



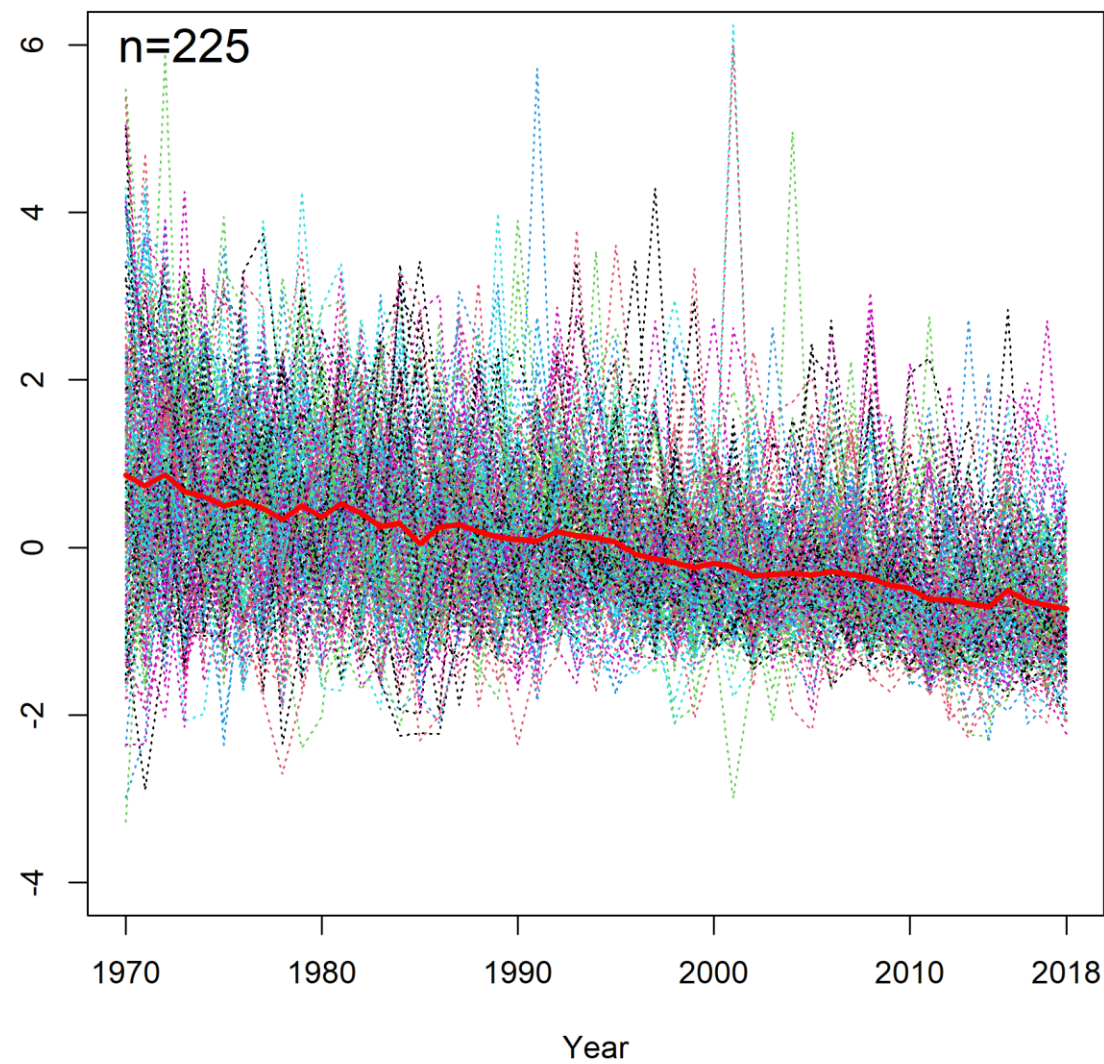
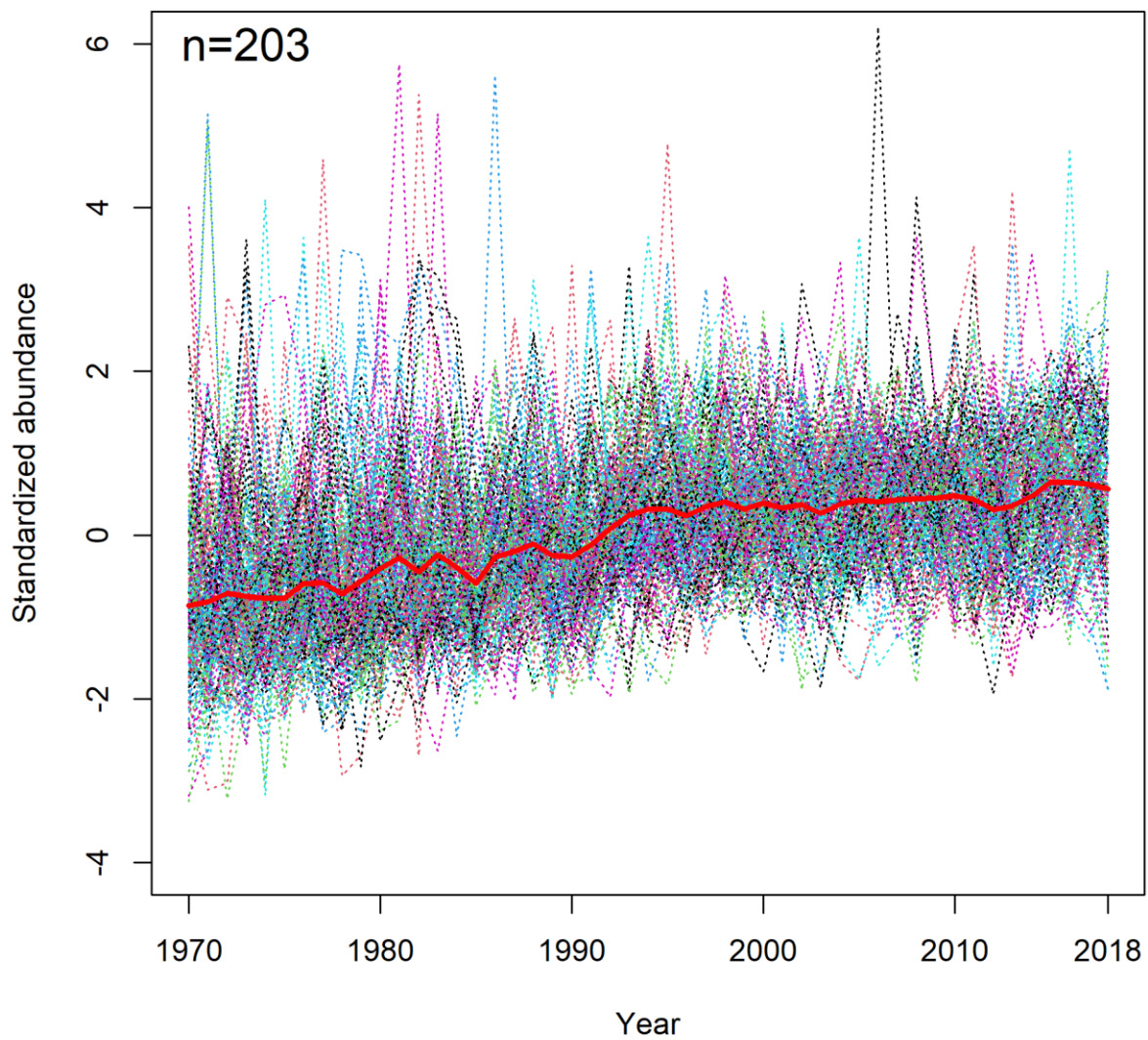
(1)



(2)



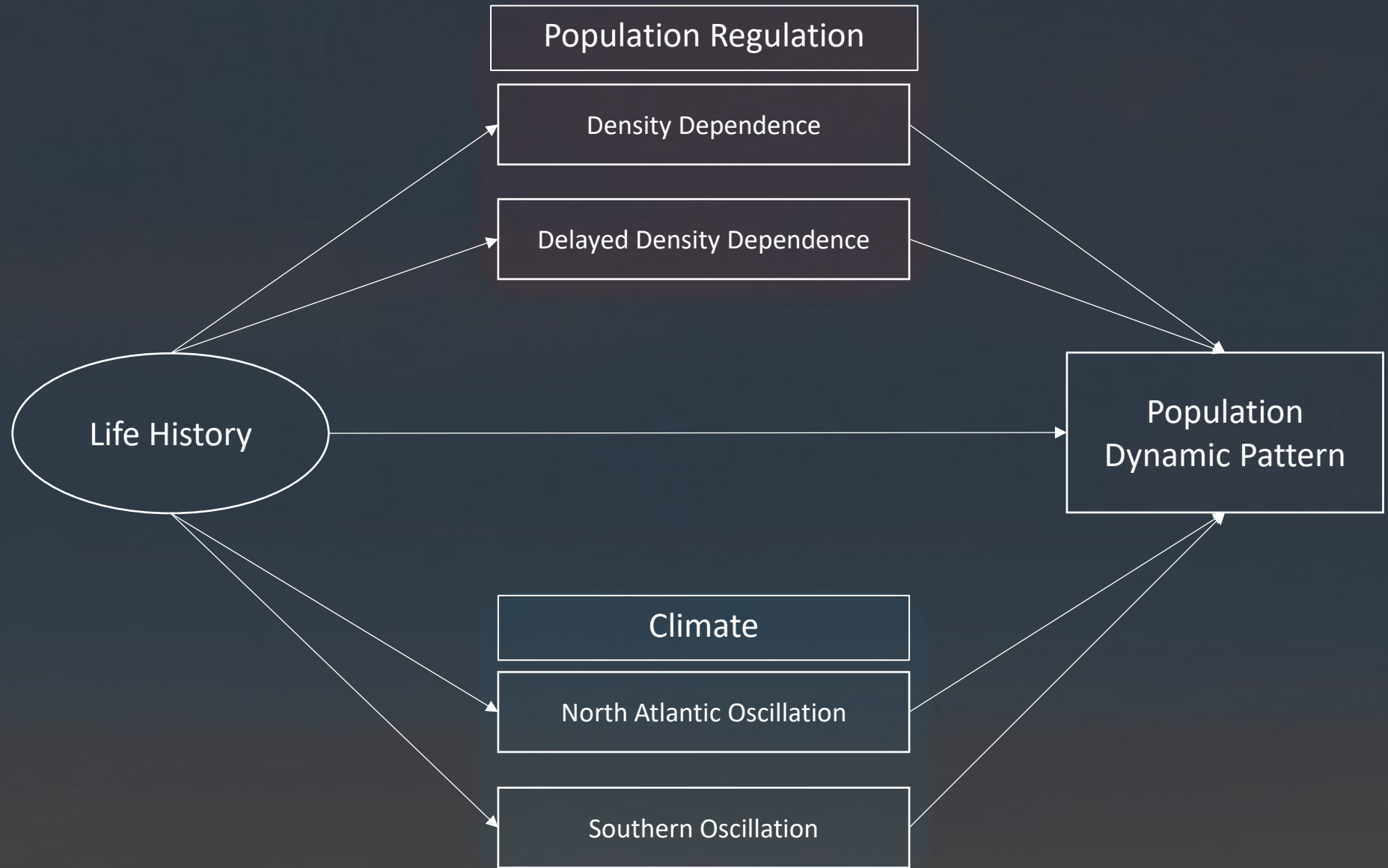
Temporal trends of North American breeding birds



Question 2

Why is
North American avifauna
declining?





Life history traits

The amniote life-history database

(Myhrvold *et al.*, 2015)

Body mass

The body mass of an adult individual in grams.

Longevity

The lifespan of an individual in years.

Annual clutches

The number of clutches produced per year.

Clutch size

The size of the clutch.

Incubation

The time between when the egg is laid and when it hatches in days.

Fledging age

The time it takes for the bird to be capable of flight (or for some species, leaving the nest) in days.

Avian hand-wing index

(Sheard *et al.*, 2020)

Range size

Number of 1° x 1° grid cells overlapping breeding/resident range.

Migration

BirdLife International migration score. Four levels: altitudinal migrant, full migrant, nomadic, and not a migrant.

Territoriality

Territoriality scores from Tobias *et al.*, 2016.
Three levels: none, weak (seasonal), and strong (year-round).

Diet

Eight diet categories: plants, nectar, seeds, fruit, invertebrates, vertebrates, scavenger, and omnivore.

Habitat

Habitat scores from Tobias *et al.*, 2016.
Three levels: dense, semi-open, and open.

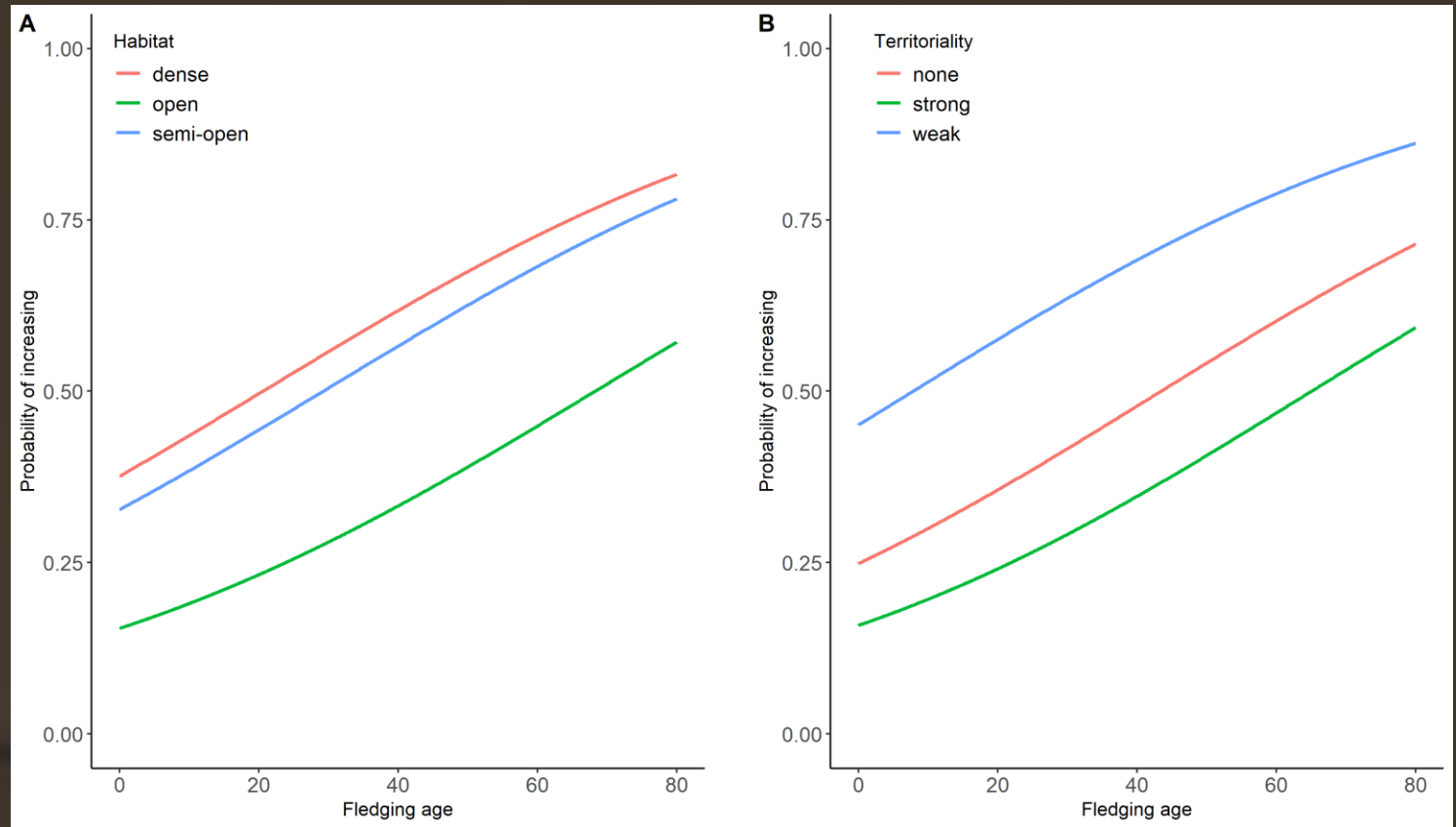


Bayesian phylogenetic logistic regression on life history traits

Variable	Coefficient	SE	LLM	ULM	$\hat{R}$
Intercept	1.3	1.72	-1.98	4.77	1.01
Clutch size	-0.12	0.25	-0.56	0.32	1.01
Fledging age*	0.59	0.33	0.01	1.23	1.01
Longevity	-0.14	0.26	-0.64	0.35	1.01
Territoriality-strong	-0.14	0.66	-1.39	1.04	1.01
Territoriality-weak*	1.34	0.48	0.47	2.26	1.01
Habitat-semi-open	0.01	0.45	-0.84	0.86	1
Habitat-open*	-1.31	0.55	-2.32	-0.36	1.01
Diet-invertebrates	-1.69	1.65	-5.04	1.44	1.01
Diet-nectar	-0.32	1.95	-4.19	3.47	1
Diet-omnivore	-2.24	1.67	-5.64	0.95	1
Diet-plants	-0.97	1.8	-4.56	2.5	1.01
Diet-scav	3.61	3.46	-2.31	11.21	1
Diet-seeds	-3.14	1.73	-6.65	0.16	1
Diet-vertebrates	-1.96	1.72	-5.44	1.3	1

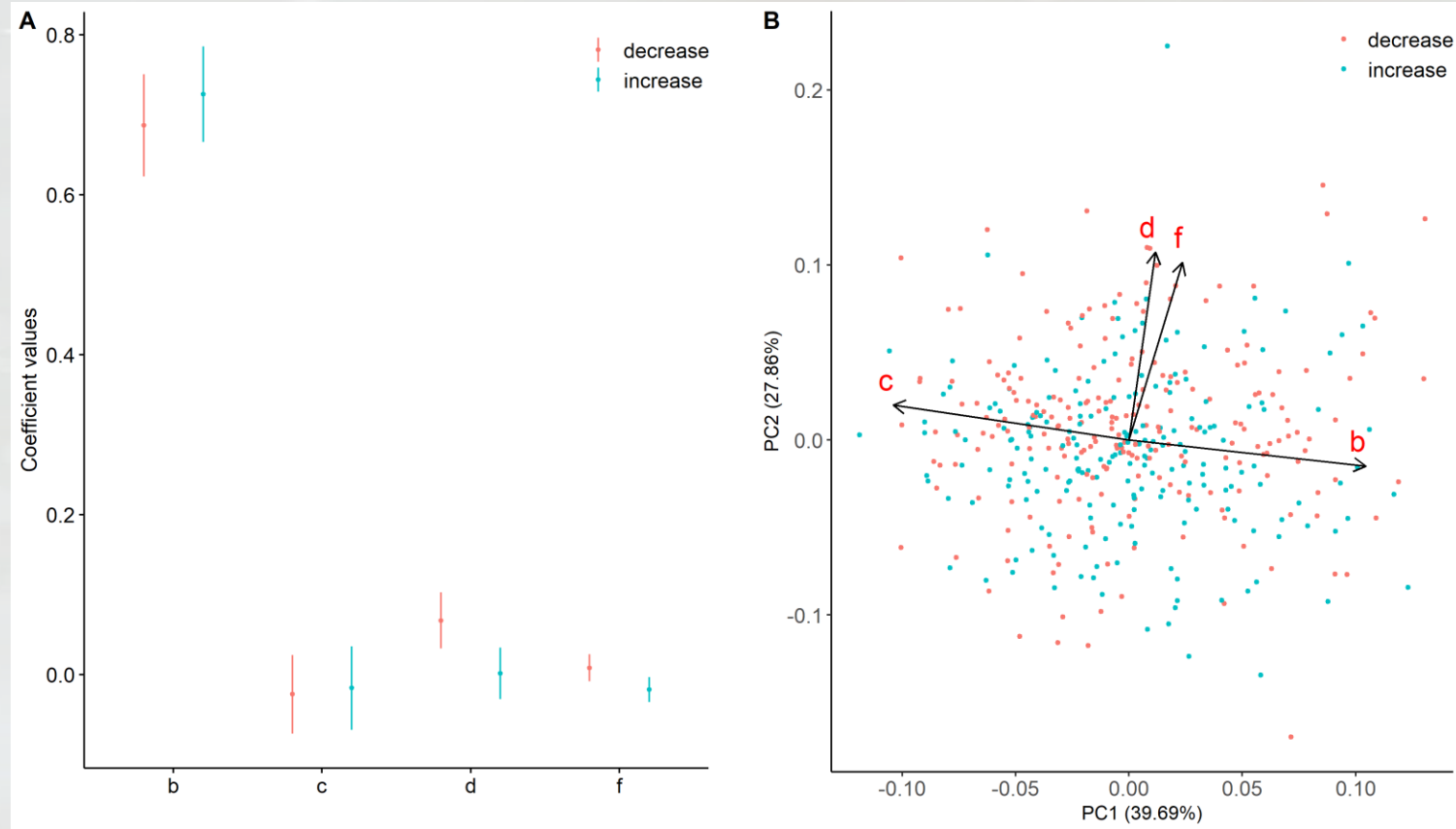


Bayesian phylogenetic logistic regression on life history traits

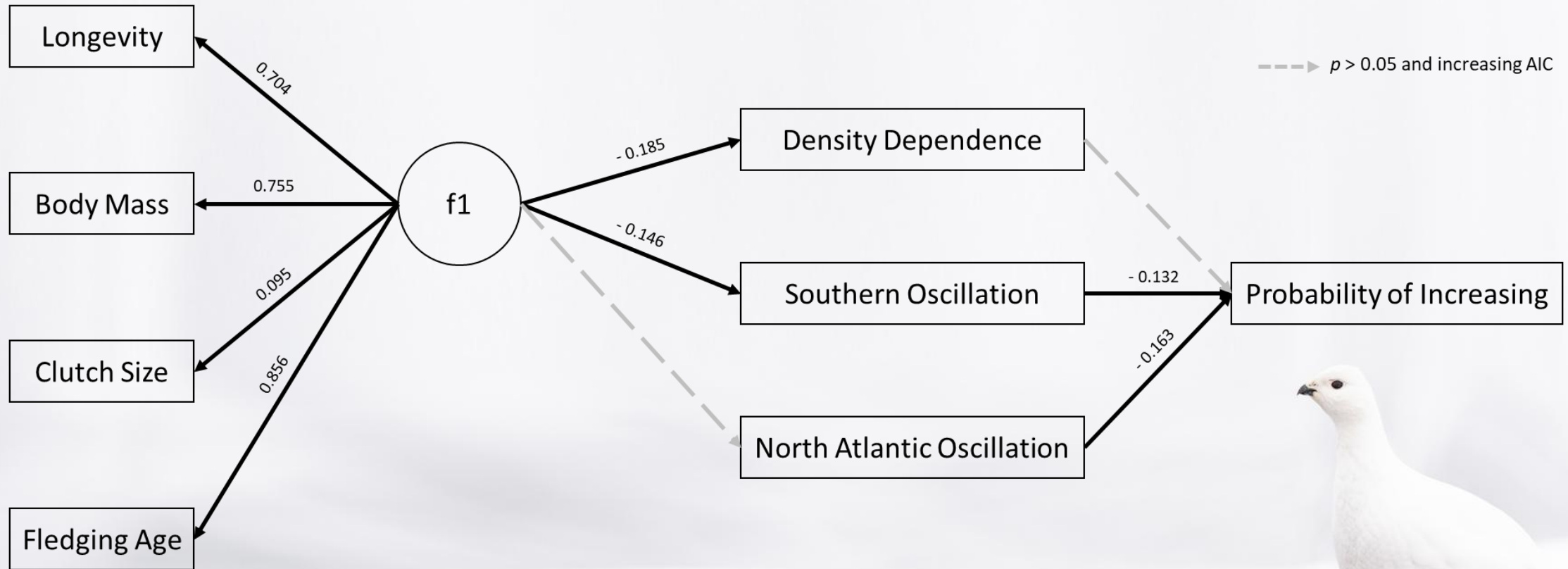


Gompertz state space population growth models

$$X_t = a + bX_{t-1} + cX_{t-2} + dNAOI_{t-1} + fSOI_{t-1} + e_t$$
$$Y_t = X_t + u_t$$



Structural equation models with nested random effect of phylogeny

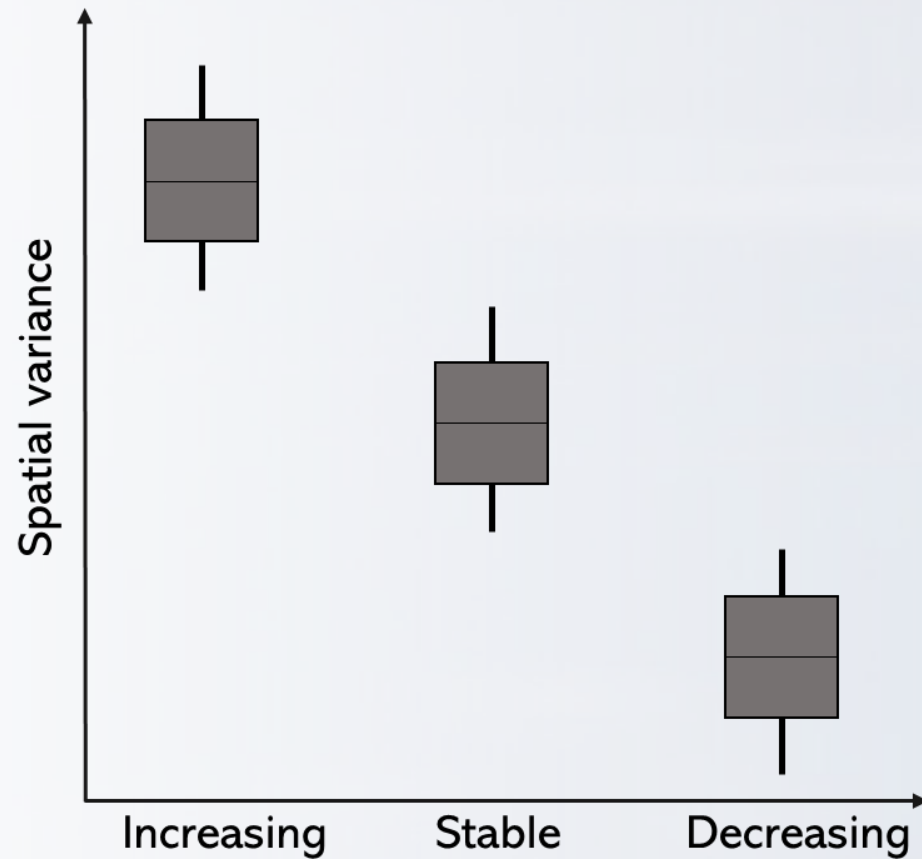


Population level
Spatial variation



Hypothesis testing

Species with increasing or stable population trends would have greater spatial variability in the population responses to climatic changes than the species with decreasing population trends.



159 species
11796 time series
141552 models



n=250

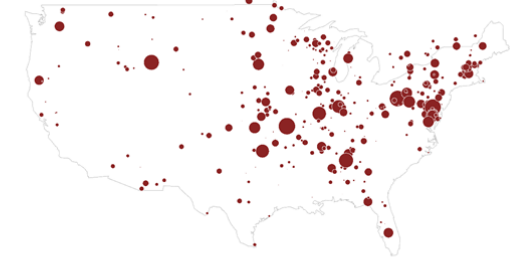
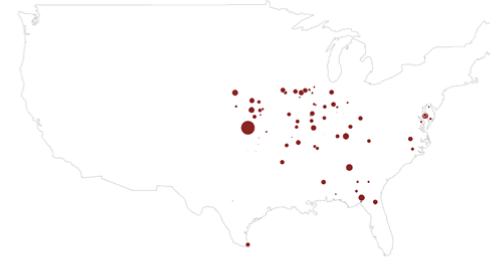
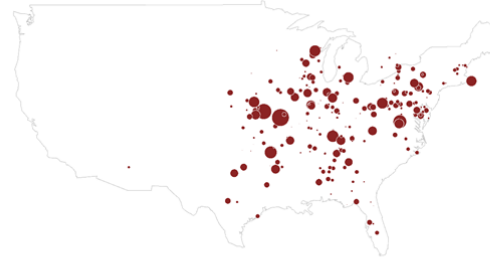


n=86

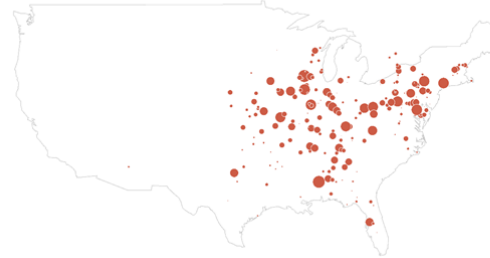


n=353

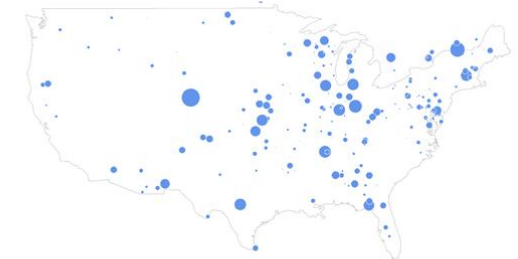
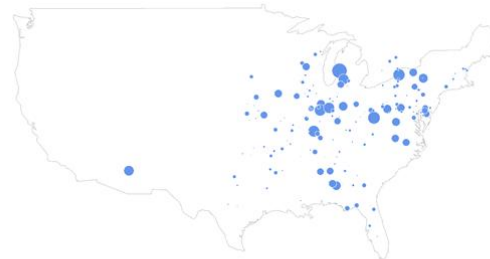
Density dependence



Delayed density dependence



Precipitation



Temperature



Take-home message

DTW-Kmeans clustering revealed two temporal dynamic trends of North American breeding bird populations.

Breeding birds having younger fledging age may be at a greater risk of long-term population declines.

Life history traits determined the strength of density dependence and the response levels of birds to SOI.

Life-history traits indirectly affected population trends by mediating the effects of climate.



Acknowledgements

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Questions?